Vitaran Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

CHEMISTRY

Class – 8

VitaranEdutech Pvt. Ltd

Hyderabad, Telangana – 72

Table of Contents

S.No	Chapter Name	Page. No.
1	Synthetic Fibres and Plastics	1-6
2	Matter and its Classification	7-14
3	Metals and Non-Metals	15-23
4	Solutions	24-26
5	Coal and Petroleum	27-30
6	Combustion and Flame	31-34

1. Synthetic Fibres and Plastics

Solutions

Worksheet – 1

1. **Correct Answer: A) Glucose**

   Explanation: Cellulose is a natural polymer made up of many glucose units joined together by glycosidic bonds.

2. **Correct Answer: B) Cellulose**

   Explanation: Cellulose occurs naturally in plants. Rayon is semi-synthetic, while polystyrene and polyester are synthetic polymers.

3. **Correct Answer: D) Protein**

   Explanation: Proteins are polymers formed by linking many amino acids through peptide bonds.

4. **Correct Answer: B) Ethene**

   Explanation: Ethene is a monomer that polymerizes to form polyethylene. The others are polymers.

5. **Correct Answer: B) Terylene**

   Explanation: Terylene (a polyester fibre) is synthetic and does not wrinkle easily, unlike natural fibres.

6. **Correct Answer: B) Cotton**

   Explanation: Cotton is a natural fibre with high cellulose content, which absorbs water easily.

7. **Correct Answer: A) Nylon**

   Explanation: Nylon is known for its high tensile strength, making it one of the strongest synthetic fibres.

8. **Correct Answer: B) Polyethylene terephthalate**

   Explanation: PET is a type of polyester widely used in making plastic bottles and fibres.

9. **Correct Answer: A) Polyester**

   Explanation: Terylene is a synthetic polyester fibre.

10. **Correct Answer: A) Cotton clothes**

    Explanation: Cotton clothes take much longer to decompose compared to vegetable peels, wood, and cow dung, which break down faster naturally.

Worksheet-2

1. **Correct Answer: B) A large molecule made by combining many monomers**

   Explanation: A polymer is formed when many small molecules called monomers chemically join together to form a long-chain, high molecular weight molecule.

2. **Correct Answer: B) Polymerisation**

   Explanation: Polymerisation is the chemical process in which monomers link together to form polymers. Condensation is only a *type* of polymerisation, not the general term.

3. **Correct Answer: C) Polythene**

   Explanation: Thermoplastics soften on heating and can be remoulded. Polythene can be melted and reshaped. Bakelite, melamine, and epoxy are thermosetting plastics.

4. **Correct Answer: C) Bakelite**

   Explanation: Bakelite is a thermosetting plastic. Once it hardens, it cannot be softened or reshaped by heating.

5. **Correct Answer: B) Adipic acid and hexamethylene diamine**

   Explanation: Nylon-6,6 is formed by condensation polymerisation between adipic acid and hexamethylene diamine.

6. **Correct Answer: B) Water bottles**

Explanation: PET (Polyethylene Terephthalate) is lightweight, strong, and safe for storing liquids, so it is widely used for bottles and food containers.

7. **Correct Answer: C) It is obtained by chemical treatment of cellulose**

Explanation: Rayon is made from natural cellulose (wood pulp) but processed chemically, so it is called semi-synthetic.

8. **Correct Answer: B) Chemical and heat resistance**

Explanation: Teflon has high resistance to heat and chemicals and a slippery surface, which prevents food from sticking.

9. **Correct Answer: C) Cotton cloth**

Explanation: Cotton is a natural fibre and can be broken down by microorganisms. Plastics, nylon, and PVC are non-biodegradable.

10. **Correct Answer: B) PET**

Explanation: Recycling code 1 represents PET (Polyethylene Terephthalate), commonly used in water and soft drink bottles.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: D) Protein**

Explanation: Proteins are natural polymers made of many amino acids joined together and are present in muscles, skin, enzymes, etc. Glucose is a monomer, and bone/cell are structures, not polymers.

2. **Correct Answer: C) Melamine**

Explanation: Melamine is a thermosetting plastic that is heat and flame resistant, so it is used in kitchenware and electrical fittings.

3. **Correct Answer: C) Non-biodegradable**

Explanation: Being non-biodegradable is a disadvantage, as synthetic fibres do not decompose easily and cause pollution.

4. **Correct Answer: B) Teflon**

Explanation: Teflon (PTFE) has a slippery surface and high heat resistance, making it ideal for non-stick cookware.

5. **Correct Answer: C) Thermosetting plastic**

Explanation: Melamine hardens permanently when heated and cannot be remoulded, which is a property of thermosetting plastics.

6. **Correct Answer: C) Nylon**

Explanation: Nylon is a man-made synthetic fibre, while jute, cotton, and wool are natural fibres.

7. **Correct Answer: A) Bakelite**

Explanation: Bakelite is a poor conductor of heat and electricity, so it prevents burns when used as utensil handles.

8. **Correct Answer: B) PET**

Explanation: PET (Polyethylene Terephthalate) is commonly used to make food and kitchen containers because it is lightweight and safe.

9. **Correct Answer: D) Fibre**

Explanation: Nylon is mainly classified as a synthetic fibre, not a plastic used for moulding objects.

10. **Correct Answer: A) Thermoplastic**

Explanation: PVC softens on heating and can be reshaped, so it is a thermoplastic.

11. **Correct Answer: D) Bottles**

Explanation: Bottles are usually made from PET, not nylon. Tents, socks, and ropes are commonly made of nylon.

12. **Correct Answer: B) Polythene**

Explanation: Polythene is a plastic, while cotton and wood pulp are natural materials.

13. **Correct Answer: A) Acrylic**

Explanation: Acrylic is called artificial wool because it looks and feels similar to wool.

14. **Correct Answer: B) Glucose**

Explanation: Cellulose is a natural polymer made of many glucose molecules joined together.

15. **Correct Answer: A) Amino acids**

Explanation: Proteins are polymers whose monomer units are amino acids.

16. **Correct Answer: A) Rayon wire**

Explanation: Nylon thread is stronger and more elastic than rayon but not stronger than steel.

17. **Correct Answer: D) Bed sheets**

Explanation: Mixing rayon and cotton produces soft, comfortable fabrics used in bed sheets and clothing.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

18. **Correct Answer: A) Starch, C) Cellulose, D) Protein**

Explanation:

- Starch, cellulose, and proteins are natural polymers found in plants and animals.
- Nylon is a synthetic (man-made) polymer, not natural.

19. **Correct Answer: A) Melamine, C) Bakelite**

Explanation:

- Thermosetting plastics harden permanently on heating and cannot be remoulded.
- Melamine and Bakelite are thermosetting plastics.
- Teflon and polythene are thermoplastics (they soften on heating).

20. **Correct Answer: A) It is a synthetic fibre, B) It is water-resistant, D) It is strong and elastic**

Explanation:

- Nylon is man-made (synthetic), does not absorb much water, and is strong and elastic.

21. **Correct Answer: A) High durability, B) Quick drying, D) Resistant to insects**

Explanation:

- Synthetic fibres are strong, long-lasting, dry quickly, and are not attacked by insects.

22. **Correct Answer: A) It is chemically inert, B) It is heat-resistant, D) It is used for non-stick coating**

Explanation:

- Teflon (PTFE) does not react easily with chemicals (chemically inert), can withstand high temperatures, and is used for non-stick cookware.

23. **Correct Answer: A) Reduce, B) Reuse, C) Repair, D) Recycle**

Explanation:

- The 5R principle includes Refuse, Reduce, Reuse, Repair, and Recycle.

ASSERTION AND REASON TYPE

24. **Correct Answer: A) Both A and R are true, and R is the correct explanation of A**

Explanation: Plastics are polymers, which are made when many monomers (small molecules) chemically join together. The reason correctly explains how plastics are formed.

25. **Correct Answer: A) Both A and R are true, and R is the correct explanation of A**

Explanation: Bakelite is a thermosetting plastic. Its cross-linked network structure becomes rigid when set, so it cannot be softened or reshaped again.

26. **Correct Answer: A) Both A and R are true, and R is the correct explanation of A**

Explanation: Rayon comes from natural cellulose (wood pulp) but is processed chemically, so it is called semi-synthetic. The reason directly explains the assertion.

27. **Correct Answer: C) Assertion is true, but Reason is false**

Explanation: Nylon is strong and durable, but it is synthetic (man-made), not a natural fibre from plants. Therefore, the reason is incorrect.

MATRIX MATCHING TYPE

28. Correct Answer: 1–d, 2–c, 3–b, 4–a

1. Correct Answers: d) Water bottles, soft drink bottles

Explanation: PET is the most common thermoplastic polymer resin. It is chosen for beverages because it is lightweight, strong, and provides a good barrier against gas (keeping the carbonation in soft drinks) and moisture.

2. Correct Answers: c) Milk jugs, shampoo bottles

Explanation: HDPE has a high strength-to-density ratio. It is sturdier and more opaque than PET, making it ideal for heavier containers like milk jugs and chemical-resistant bottles for shampoo or laundry detergent

3. Correct Answers: b) Pipes, wire insulation

Explanation: PVC is a high-strength thermoplastic material. Because it is extremely durable, biologically and chemically resistant, and can be made either rigid or flexible, it is the standard for plumbing pipes and electrical cable insulation.

4. Correct Answers: a) Plastic bags, wraps

Explanation: LDPE has a more "branched" molecular structure than HDPE, which makes it thinner and more flexible. This flexibility is why it's used for items that need to stretch or fold, such as grocery bags and plastic films/wraps.

29. Correct Answer: 1–C, 2–A, 3–B, 4–D

1. Correct Answers: C) Plant fibre

Explanation: Cotton comes from the cotton plant, so it is a natural plant fibre.

2. Correct Answers: A) Semi-synthetic fibre

Explanation: Rayon is made from natural cellulose but processed chemically, so it is semi-synthetic.

3. Correct Answers: B) Synthetic fibre

Explanation: Nylon is man-made using chemicals, so it is a synthetic fibre.

4. Correct Answers: D) Animal fibre

Explanation: Wool is obtained from sheep and other animals, so it is an animal fibre.

COMPREHENSION TYPE

30. Correct Answer: B) Polymers of high molecular weight

Explanation: Plastics are made of long chains of repeating units (monomers), which gives them a high molecular weight and special properties like strength and flexibility.

31. Correct Answer: B) Thermoplastics

Explanation: Thermoplastics soften when heated and can be moulded again and again (e.g., polythene, PVC). Thermosetting plastics cannot be reshaped once set.

32. Correct Answer: B) Have cross-linked structures

Explanation: Thermosetting plastics form a rigid, cross-linked network when heated, which makes them hard and irreversible in shape.

33. Correct Answer: B) Identify plastic types for sorting

Explanation: The recycling code shows what kind of plastic the product is made from, helping workers and machines sort it correctly.

34. Correct Answer: B) Prevents contamination in recycling

Explanation: Mixing different types of plastics can spoil the recycling process, so proper sorting keeps the recycled material clean and useful.

35. Correct Answer: C) Recycling and plastic identification code

Explanation: The triangle with arrows is the recycling symbol that contains a number to identify the type of plastic used.

INTEGER TYPE

36. Correct Answer: 2

Explanation: Fibres are classified into two main types based on their source:

- Natural fibres (cotton, wool, jute, silk)
- Synthetic fibres (nylon, polyester, acrylic)

(Some books also mention semi-synthetic fibres like rayon under man-made fibres, but the main classification is two.)

37. Correct Answer: 2

Explanation: Nylon-6,6 is formed by the reaction of two different monomers:

- Adipic acid
- Hexamethylene diamine

38. Correct Answer: 7

Explanation: Plastics are identified using 7 standard recycling codes, numbered 1 to 7 (e.g., 1 = PET, 2 = HDPE, 3 = PVC, etc.).

39. Correct Answer: 2

Explanation: Based on how they behave when heated, plastics are classified into:

- Thermoplastics (can be softened and reshaped)
- Thermosetting plastics (become permanently hard)

Previous Year Questions

1. Correct Answer: C

Explanation: Materials (iii) and (vi) have polymer chains with crosslinks which help them to change shape on heating.

While it is true that Bakelite and Melamine have cross-linked chains, this structure makes them Thermosetting plastics. Once they are molded and set, they do not soften or change shape upon reheating; instead, they char or burn.

2. Correct Answer: B

Explanation: Polythene and PVC (Polyvinyl Chloride) are actually thermoplastics, not thermosetting plastics. Thermoplastics can be softened by heating and reshaped multiple times.

3. (a) Fishing nets — (iii) Nylon: Nylon is used because of its high tensile strength, elasticity, and water resistance.

(b) Blanket — (iv) Acrylic: As mentioned above, acrylic is "synthetic wool" and is ideal for warm clothing and blankets.

(c) Squeeze bottles — (v) Polythene: Low-density polythene is flexible and durable, making it perfect for squeezable containers.

(d) Plumbing materials — (ii) PVC: PVC is rigid and chemically resistant, making it the standard for pipes and plumbing.

(e) Fire resistant fabric — (i) Melamine: Melamine is a thermosetting plastic that can tolerate high heat and is used in firemen's uniforms and floor tiles.

4. Correct Answer: C (Statement 1 is true but statement 2 is false)

Explanation: Statement 1 (True): Melamine is indeed a thermosetting plastic. It is versatile and used for making floor tiles, kitchenware (handles of utensils), and electrical switches because it resists fire and tolerates heat better than other plastics.

Statement 2 (False): Melamine is a poor conductor (an insulator) of heat and electricity. This property is exactly why it is used to make handles for utensils (so you don't burn your hand) and electrical switches (to prevent electric shocks)

5. Correct Answer: A (Polyester, nylon, Teflon, and rayon)

Explanation:

1. Polyester fabrics do not wrinkle easily.

2. Nylon is used for making parachutes and stockings due to its strength.
3. Teflon is used to make non-stick cookwares because oil and water do not stick to it.
4. Rayon is used to make carpets (often mixed with wool) and upholstery.

6. Correct Answer: A (i), (ii) and (v) logic check)

Explanation: (i) Polythene & (v) PVC are Thermoplastics.

(ii) Bakelite & (iii) Teflon... actually, Teflon is a thermoplastic, but Bakelite is definitely a Thermosetting plastic.

(iv) Polycot is a blended fiber (polyester + cotton).

7. Correct Answer: A

Explanation:

- X is PVC: A thermoplastic used widely for the soles of shoes due to its durability and flexibility.
- Y is Bakelite: A thermosetting plastic used for electrical switches and plugs because it is an excellent insulator and does not soften when heated.

8. Correct Answer: B (Rayon)

Explanation: The question describes a fiber obtained from a natural polymer (cellulose) that "shines like silk" and is used to make carpets (as shown in the image). This is the classic definition of Rayon, also known as artificial silk.

9. Correct Answer: B (Nylon)

Explanation: Synthetic fibers like Nylon absorb very little water compared to natural fibers. Therefore, when weighed after soaking, Nylon will show the least difference from its original dry weight. Cotton and wool are highly absorbent and would show significant weight increases.

2. Matter and Its Classification

Solutions

Worksheet – 1

1. **Correct Answer: D) Five.**

   Explanation: Matter is commonly classified into five states : solid, liquid, and gas, plasma and Bose–Einstein condensate .

2. **Correct Answer: C) Solid**

   Explanation: In solids, molecules are closely packed and held in fixed positions. This gives solids a definite shape and volume.

3. **Correct Answer: C) Definite volume but no definite shape**

   Explanation: Liquids take the shape of the container they are in but maintain a fixed volume because the molecules are close but can move past each other.

4. **Correct Answer: C) Have no definite volume**

   Explanation: Gas molecules are far apart and move freely. Gases have neither a definite shape nor a definite volume; they expand to fill the container.

5. **Correct Answer: C) Gas**

   Explanation: Gas molecules are widely spaced compared to solids and liquids, so intermolecular space is maximum in gases.

6. **Correct Answer: A) Solid**

   Explanation: Solids are rigid because their molecules are fixed in position and cannot move freely.

7. **Correct Answer: C) Vibration about fixed positions**

   Explanation: Molecules in solids cannot move freely; they only vibrate around their fixed positions.

8. **Correct Answer: C) Can move freely**

   Explanation: Molecules in liquids are less tightly packed than solids and can move past one another, which allows liquids to flow.

9. **Correct Answer: C) Gas**

   Explanation: Gas molecules are far apart and have very weak intermolecular forces, so they move independently.

10. **Correct Answer: C) Stars**

    Explanation: Plasma is an ionized gas found in extremely hot environments, like the Sun and other stars.

Worksheet-2

1. **Correct Answer: C) Melting point**

   Explanation: The melting point is the temperature at which a solid absorbs heat and turns into a liquid. For example, ice melts at 0°C.

2. **Correct Answer: B) Boiling or Vapourisation**

   Explanation: When a liquid gains enough heat, it changes to gas. If this happens throughout the liquid, it is called boiling; if it happens at the surface, it is evaporation.

3. **Correct Answer: C) 100°C**

   Explanation: Condensation occurs at the boiling point of water (100°C or 373 K) when water vapor loses heat and turns back into liquid.

4. **Correct Answer: B) Solidification or Freezing**

   Explanation: Freezing occurs when a liquid loses heat and becomes solid. For example, water freezes at 0°C.

5. **Correct Answer: B) A gas is cooled**

   Explanation: Condensation is the process where a gas loses heat and changes into a liquid.

6. **Correct Answer: B) Sublimation**

   Explanation: Sublimation occurs when solids like iodine or dry ice turn directly into gas without melting.

7. **Correct Answer: A) It absorbs heat**

   Explanation: Melting requires heat energy, which breaks the bonds between ice molecules, turning it into water.

8. **Correct Answer: B) Gas can convert directly into solid**

   Explanation: This process is called deposition. Other statements are incorrect because solids can sometimes sublimate and melting occurs on heating, not cooling.

9. **Correct Answer: B) 373 K**

   Explanation: Boiling point of water is 100°C, which is equivalent to 373 K.

10. **Correct Answer: B) Heat is released**

    Explanation: During freezing, a liquid loses heat to its surroundings, allowing molecules to settle into a solid structure.

Worksheet-3

1. **Correct Answer: C) Copper (Cu)**

   Explanation: A pure substance contains only one type of particle with definite properties. Copper is an element, so it is pure. Milk, tap water, and honey are mixtures.

2. **Correct Answer: A) Water**

   Explanation: A compound has elements in a fixed ratio. Water (H_2O) always contains hydrogen and oxygen in a 2:1 ratio by atoms (or 1:8 by weight).

3. **Correct Answer: C) Chemical reactions**

   Explanation: Compounds are chemically bonded. To separate them into elements, chemical reactions are required, not physical methods like filtration or evaporation.

4. **Correct Answer: B) Nitrogen (N)**

   Explanation: Nitrogen is a gas at room temperature. Mercury is a liquid, bromine is a liquid, and sulfur is a solid.

5. **Correct Answer: B) Radioactive elements**

   Explanation: Elements like Uranium and Radium emit radiation naturally and are called radioactive elements.

6. **Correct Answer: A) Graphite has weak bonds between layers, diamond has a 3D network structure**

   Explanation: Diamond has a strong 3D covalent structure, making it hard. Graphite has layers held together by weak forces, making it soft and slippery.

7. **Correct Answer: B) It contains dissolved salts and air**

   Explanation: Tap water is a mixture of water, salts, and gases; hence it is not pure.

8. **Correct Answer: C) Retain their individual properties**

   Explanation: Mixture components keep their own properties and are not chemically bonded.

9. **Correct Answer: B) Uniformly distributed**

   Explanation: In homogeneous mixtures (like sugar in water), the composition is uniform, and you cannot see separate parts.

10. **Correct Answer: C) Sand in water**

    Explanation: In sand in water, the components are not uniformly distributed and can be seen separately. Air and sugar/salt solutions are homogeneous.

Worksheet-4

1. **Correct Answer: A) Threshing → Winnowing → Sieving**

   Explanation:

   - Threshing separates grains from stalks.
   - Winnowing removes lighter husk using wind.
   - Sieving separates particles of different sizes.

2. **Correct Answer: A) The components are of different colors and easily visible**

   Explanation: Handpicking works when impurities are large and can be easily seen and removed by hand.

3. **Correct Answer: B) Mining industries**

   Explanation: Magnetic separation removes magnetic materials like iron from ores in mining.

4. **Correct Answer: B) Filtration → Evaporation**

   Explanation:

   - Filtration separates insoluble sand.
   - Evaporation removes water to obtain salt.

5. **Correct Answer: C) Refining crude oil**

   Explanation: It separates liquids with close boiling points into useful fuels.

6. **Correct Answer: A) Evaporates first**

   Explanation: The substance with lower boiling point vaporizes earlier and is collected first.

7. **Correct Answer: A) It produces pure crystals**

   Explanation: Crystallisation gives pure solid crystals, while simple evaporation may leave impurities.

8. **Correct Answer: C) Difference in density**

   Explanation: Heavier particles settle outward faster when spun at high speed.

9. **Correct Answer: B) Physical separation methods**

   Explanation: They separate substances without any chemical change

10. **Correct Answer: A) Water purification**

    Explanation: Coagulation helps small suspended particles clump together for easy removal.

Worksheet-5

1. **Correct Answer: B) It changes directly into vapour**

   Explanation: In sublimation, a solid converts directly into vapour without becoming liquid.

2. **Correct Answer: A) Remove dissolved gases from liquids by heating**

   Explanation: Heating reduces gas solubility, so dissolved gases escape from the liquid.

3. **Correct Answer: A) Industrial boilers and water pipelines**

   Explanation: Removing dissolved gases prevents corrosion and damage in pipelines and boilers.

4. **Correct Answer: B) Solubility in a liquid**

   Explanation: A specific gas dissolves in a liquid solvent and is removed from the mixture.

5. **Correct Answer: B) Difference in density**

   Explanation: Lighter gas molecules diffuse faster than heavier ones.

6. **Correct Answer: B) Cooled and compressed until it liquefies**

   Explanation: The gases are liquefied first, then separated based on different boiling points.

7. **Correct Answer: A) Different components move at different rates depending on their solubility and adsorption**

   Explanation: Substances travel at different speeds due to different attractions to the paper and solvent.

8. **Correct Answer: B) Filtration after dissolving in water**

   Explanation: Salt dissolves in water, sand does not; filtration separates sand.

9. **Correct Answer: B) Sand**

   Explanation: Sand is insoluble and stays as residue on the filter paper.

10. **Correct Answer: B) Magnetic separation → Dissolution and filtration → Evaporation**

    Explanation:
    Magnet removes iron
    Salt dissolves in water and sand is filtered
    Evaporation recovers salt

Competitive corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: B) Mercury**

   Explanation: Mercury is the only metal that is liquid at room temperature (~25°C). Chlorine and oxygen are gases, and carbon is solid.

2. **Correct Answer: C) Sulfur**

   Explanation: Sulfur is a non-metal and solid at room temperature. Helium and neon are gases, bromine is a liquid.

3. **Correct Answer: B) Table salt (NaCl)**

   Explanation: Pure substances like NaCl have a fixed composition and definite properties. Milk, orange juice, and soft drinks are mixtures.

4. **Correct Answer: B) Variable**

   Explanation: Mixtures can have variable proportions of their components, unlike compounds which have a fixed ratio.

5. **Correct Answer: A) Filtration**

   Explanation: Physical methods like filtration, evaporation, or distillation can separate mixtures. Combustion and oxidation are chemical changes.

6. **Correct Answer: B) Fixed**

   Explanation: Compounds always have elements in a fixed proportion by mass. For example, water always has 2 hydrogen atoms and 1 oxygen atom.

7. **Correct Answer: C) Chemical reaction with energy change**

   Explanation: Compounds form via chemical reactions, often with energy absorbed or released.

8. **Correct Answer: C) Air**

   Explanation: Matter has mass and occupies space. Air is matter, but heat, light, and sound are forms of energy, not matter.

9. **Correct Answer: B) Ionized gas with free electrons and ions**

   Explanation: Plasma is a high-energy state where electrons are separated from nuclei, forming ions. Found in stars and lightning.

10. **Correct Answer: C) Very low temperature near absolute zero**

    Explanation: Bose–Einstein Condensates occur when atoms are cooled extremely close to 0 K, causing them to occupy a single quantum state.

11. **Correct Answer: B) Can flow**

    Explanation: Fluids take the shape of their container and can flow due to freely moving particles.

12. **Correct Answer: C) Gases**

    Explanation: Gases expand the most on heating because particles are far apart and move freely.

13. **Correct Answer: B) Definite mass**

    Explanation: All states of matter have mass and occupy space, but shape and density vary.

14. **Correct Answer: C) Have neither definite shape nor definite volume**

    Explanation: Gas molecules move freely and spread to fill the container.

15. **Correct Answer: B) Eric Cornell and Carl Wieman**

    Explanation: They first created BEC in 1995 using ultra-cold rubidium atoms.

16. **Correct Answer: A) Salt dissolves in water forming a homogeneous mixture**

    Explanation: Salt water is a solution (homogeneous mixture), which can be separated physically via distillation.

17. **Correct Answer: B) It is a solid–liquid mixture with suspended particles**

    Explanation: Filtration removes solid particles from liquids, making the water safe.

18. **Correct Answer: B) It contains oxygen, nitrogen, and other gases mixed physically**

Explanation: Air is a homogeneous mixture of gases; the components are not chemically bonded.

19. **Correct Answer: B) Cement and construction**

Explanation: Limestone is a key raw material in cement, mortar, and construction due to its chemical properties.

20. **Correct Answer: C) An excellent insulator**

Explanation: Mica does not conduct electricity and can withstand heat, making it ideal for electrical insulation.

21. **Correct Answer: C) Exactly 10 g**

Explanation: By the law of conservation of mass, mass is neither created nor destroyed in a closed system.

22. **Correct Answer: C) Increases**

Explanation: Higher temperature → particles move faster → kinetic energy increases.

23. **Correct Answer: B) Are tightly packed and move very little**

Explanation**:** Solid particles have very little freedom of movement, so diffusion is slow.

24. **Correct Answer: C) Particles gain kinetic energy and move faster**

Explanation: Heating increases particle motion, causing gas to expand.

25. **Correct Answer: B) Evaporation**

Explanation: Water evaporates, leaving solid salt behind.

26. **Correct Answer: B) Sublimation**

Explanation: Camphor changes directly from solid to vapour on heating.

27. **Correct Answer: A) Identifying pigments in plants**

Explanation: It separates colored components based on movement through paper.

28. **Correct Answer: B) Dissolving into a liquid solvent**

Explanation: The gas dissolves selectively in the solvent.

29. **Correct Answer: B) Lighter gases pass through the porous barrier more quickly than heavier gases**

Explanation: Lighter molecules move faster and diffuse more quickly.

30. **Correct Answer: A) Separating nitrogen and oxygen from air**

Explanation: Air is liquefied and gases are separated by different boiling points.

31. **Correct Answer: A) Preparing distilled water for laboratories**

Explanation: Heating removes dissolved gases from water, making it purer.

ONE OR MORE THAN OPTION CORRECT TYPE

32. **Correct Answer: A) Definite shape, B) Definite volume, C) Molecules tightly packed**

Explanation: Solids have a definite shape and volume because their molecules are tightly packed and held in fixed positions. They do not have “no intermolecular force”; in fact, strong intermolecular forces hold the molecules together.

33. **Correct Answer: A) Liquids, B) Gases, D) Plasma**

Explanation: Fluids are substances that can flow. Solids cannot flow. Both liquids, gases, and plasma are considered fluids.

34. **Correct Answer: A) More intermolecular space, B) Weaker intermolecular force, C) Ability to flow**

Explanation: Liquids have molecules less tightly packed than solids, allowing them to flow. Solids have fixed shapes, unlike liquids.

35. **Correct Answer: A) Pure ice, D) Oxygen gas**

Explanation: Pure substances like ice (H_2O) or oxygen gas have sharp melting or boiling points. Mixtures like sugar solution or salt and sand melt gradually over a range of temperatures.

36. Correct Answer: A) Melts gradually over a temperature range, C) Composition can be changed by physical methods

Explanation: Impure substances or mixtures do not have sharp melting points and their composition can be altered physically. Pure substances have a fixed composition.

37. Correct Answer: A) Salt solution, B) Air

Explanation: Homogeneous mixtures are uniform in composition. Salt solution (salt dissolved in water) and air (mixture of gases) are uniform, while sand and sugar mixture is heterogeneous.

ASSERTION AND REASON TYPE

38. Correct Answer: A) Both A and R are true, and R is the correct explanation of A

Explanation: Sugar and salt solution is a mixture, not a single pure substance. It can be separated by physical methods like evaporation, which is why the reason correctly explains the assertion.

39. Correct Answer: A) Both A and R are true, and R is the correct explanation of A

Explanation: Diamond and graphite are pure forms of carbon (elements), but their different structures give them different physical properties. The reason explains the assertion correctly.

40. Correct Answer: A) Both A and R are true, and R is the correct explanation of A

Explanation: BEC occurs near absolute zero. At these temperatures, atoms occupy the same quantum state and behave collectively as a single quantum entity, which is why the reason correctly explains the assertion.

41. Correct Answer: C) A is true, but R is false

Explanation:

Salt can be obtained by evaporation because water evaporates on heating, leaving solid salt behind.Salt is soluble in water, not insoluble. It dissolves in water first and is recovered only after the water evaporates.

MATRIX MATCHING TYPE

42. 1-b,2-a,3-d,4-c

Explanation:

- Gas: Molecules are far apart and move freely → no definite shape or volume.
- Plasma: High-energy ionized gas found in stars, lightning.
- Bose–Einstein Condensate: Ultra-cold state near absolute zero, atoms act as one quantum unit.
- Alloy: Mixture of metals, homogeneous in nature (e.g., brass, bronze).

43. 1-c,2-b,3-,4-a

Explanation:

- Filtration separates insoluble sand from salt solution.
- Sublimation: Direct change from solid → gas (e.g., iodine, dry ice).
- Water evaporates, leaving solid salt behind.
- Salt dissolves, sand remains insoluble.

COMPREHENSION TYPE

44. Correct Answer: B) Sugar

Explanation: A pure substance contains only one type of particle with definite properties. Sugar (sucrose) is a pure compound. Air, sea water, and salt solution are mixtures, so they are impure substances.

45. Correct Answer: C) Melting and boiling points

Explanation: Pure substances have sharp melting and boiling points. Impure substances or mixtures melt/boil over a range of temperatures, so this property is used to distinguish them.

46. Correct Answer: D) All of the above

Explanation:

- Distillation separates liquids based on boiling point differences.
- Sublimation separates solids that change directly to gas.
- Filtration separates solid particles from liquids.

All these are physical methods used to separate components of mixtures (impure substances).

47. Fractional distillation is used to separate:

Correct Answer: B) Liquids with close boiling points
Explanation: Fractional distillation separates liquids that have similar boiling points by repeated vaporization and condensation.

48. The fractionating column helps in:

Correct Answer: B) Repeated condensation and vaporization for better separation
Explanation: The fractionating column provides a large surface area where vapours condense and re-evaporate many times, improving separation efficiency.

49. A practical application of fractional distillation is:

Correct Answer: B) Refining crude oil into petrol and diesel
Explanation: Different components of crude oil have different (but close) boiling points and are separated into useful fuels using fractional distillation

INTEGER TYPE

50. Correct Answer: Close to 0 K (very near absolute zero)

Explanation:
A Bose–Einstein Condensate (BEC) forms when particles are cooled to extremely low temperatures near 0 Kelvin.
At this temperature, particle motion becomes minimal and atoms behave like a single quantum state.

51. Correct Answer: 5

Explanation:
When advanced states are included, matter exists in:

1. Solid
2. Liquid
3. Gas
4. Plasma
5. Bose–Einstein Condensate

52. Correct Answer: 4 scientists

Explanation:

- Prediction: Satyendra Nath Bose and Albert Einstein
- First production: Eric Cornell and Carl Wieman

53. Correct Answer: 2 types

Explanation:
Mixtures are classified into:

1. Homogeneous mixtures → uniform composition (e.g., salt solution)
2. Heterogeneous mixtures → non-uniform composition (e.g., sand and water)

54. Correct Answer: –183 °C

Explanation:
In cryogenic distillation, air is first liquefied. When the temperature is slowly raised, each component boils at its own boiling point. Oxygen boils at –183 **°C**, so it vaporizes at this temperature and can be separated from other gases like nitrogen (which boils at a lower temperature).

Previous Year Questions

55. Correct Answer: B) II and IV only

Explanation:

Statement II: *Ice at 273 K is more effective in cooling than water at 273 K.*

- This is correct. Ice absorbs latent heat of fusion while melting, so it can remove more heat from surroundings than water at the same temperature.

Statement IV: Evaporation and fusion processes involve absorption of heat.

- This is correct. Both fusion (melting) and evaporation are endothermic processes.

56. Correct Answer: B) (ii),(iii) and (iv) → colloid / (i),(v) and (vi) → suspension

Explanation: Mixtures given:
(i) Shaving cream – Colloid (air dispersed in liquid)
(ii) Clouds – Colloid (water droplets in air)
(iii) Chalk powder and water – Suspension (solid settles on standing)
(iv) Smoke – Colloid (solid in gas)
(v) Muddy water – Suspension (solid settles on standing)
(vi) Face cream – Colloid (liquid in liquid emulsion)

57. Correct Answer: C

Explanation: Separation of miscible and immiscible liquids

- Substance P boils at 475 K
- Substance Q boils at 350 K
- They are miscible

Method to separate miscible liquids: Distillation (simple or fractional)

58. Correct Answer: C) 20 g

Explanation: Since all 20 g of salt was dissolved and water is only reduced to half, none of the salt has evaporated or lost.

On evaporating all the water eventually, all 20 g of salt can be recovered.

59. Correct Answer: C) R

Explanation: We are given chromatograms:

- Food sample that does not contain any harmful components → spot does not match X, Y, Z.

Check the chromatogram: The sample with no spots at X, Y, Z positions correspond to R.

60. Correct Answer: C) III

1. Explanation: 10 g NaCl in 200 g water → solution mass = 210 g
$$\% = \frac{10}{210} \times 100 \approx 4.76\%$$
2. 15 g sugar in 160 g water → solution mass = 175 g
$$\% = \frac{15}{175} \times 100 \approx 8.57\%$$
3. 60 g $KMnO_4$ in 200 g water → solution mass = 260 g
$$\% = \frac{60}{260} \times 100 \approx 23.08\%$$
4. 20 g Na_2CO_3 in 90 g water → solution mass = 110 g
$$\% = \frac{20}{110} \times 100 \approx 18.18\%$$

3. Metals and Non-Metals

Solutions

Worksheet – 1

1. **Correct Answer: A) Malleability**

   Explanation:

   Malleability: Metals can be hammered or rolled into thin sheets without breaking. This is because the layers of metal atoms can slide over each other while maintaining metallic bonding.

2. **Correct Answer: C) Silver**

   Explanation:

   Sonority is the ability of a metal to produce a clear ringing sound when struck.

3. **Correct Answer: D) Bromine**

   Explanation: Bromine is a non-metal liquid and does not have a shiny surface. Copper, iodine, and silver show lustre.

4. **Correct Answer: C) Zinc**

   Explanation: Zinc is brittle at room temperature and becomes malleable when heated to about 100–150°C.

5. **Correct Answer: B) Hydrogen**

   Explanation: Hydrogen makes up the largest proportion of matter in the universe, mainly in stars.

6. **Correct Answer: D) Zinc**

   Explanation: Gold, silver, and platinum are lustrous and corrosion-resistant, making them suitable for ornaments. Zinc is not commonly used.

7. **Correct Answer: D) Both A and C**

   Explanation: Non-metals mostly occur as solids (like carbon, sulfur) and gases (like oxygen, nitrogen). Only one common non-metal (bromine) is liquid.

8. **Correct Answer: C) Dullness**

   Explanation: Metals are usually lustrous (shiny), not dull. They do conduct electricity, are sonorous, and are ductile.

9. **Correct Answer: D) (a) & (d)**

   Explanation: Aluminium is a good conductor of heat and has a high melting point, making it suitable for cooking utensils.

10. **Correct Answer: A) Malleable**

    Explanation:
    Metals are generally malleable, meaning they can be beaten into thin sheets. Most metals have high melting points, are usually hard (not soft), and are not brittle (they bend rather than break, except for a few like zinc).

Worksheet-2

1. **Correct Answer: B) Basic**

   Explanation: Most metal oxides react with acids to form salt and water, so they are basic in nature.
   Example:
   $CaO + 2HCl \rightarrow CaCl_2 + H_2O$

2. **Correct Answer: A) Acidic**

   Explanation: Non-metal oxides usually react with water to form acids.
   Example:
   $CO_2 + H_2O \rightarrow H_2CO_3$ (Carbonic acid)

3. **Correct Answer: A) Zn**

   Explanation: Zinc is more reactive than copper, so it can displace copper from its salt solution.
   $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

4. **Correct Answer: D) Ca > Mg > Zn > Cu**

   Explanation: Calcium is most reactive, followed by magnesium, then zinc, and copper is the least reactive among these.

5. **Correct Answer: B) Ag_2S**

   Explanation: Silver reacts with sulphur compounds in air (like H_2S) to form silver sulphide (Ag_2S), which is black in colour.

6. **Correct Answer: C) Al**

   Explanation: Aluminium oxide (Al_2O_3) reacts with both acids and bases, so it is amphoteric.

7. **Correct Answer: D) $CuSO_4$ + Fe**

   Explanation: Iron is more reactive than copper, so it displaces copper from copper sulphate.
   $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

8. **Correct Answer: A) CO**

   Explanation: Carbon monoxide (CO) is a neutral oxide because it does not show acidic or basic nature.

9. **Correct Answer: A) Aluminium**

   Explanation: Metals above hydrogen in the reactivity series can displace hydrogen from acids. Aluminium lies above hydrogen, while silver, mercury, and copper lie below it.

10. **Correct Answer: C) Hydrogen**

    Explanation: When reactive metals react with water, hydrogen gas is released.
    Example:
    $2Na + 2H_2O \rightarrow 2NaOH + H_2\uparrow$

Worksheet-3

1. **Correct Answer: B) Stainless steel**

   Explanation:

   - Stainless steel = Iron + Chromium (and sometimes Nickel, Carbon)

2. **Correct Answer: C) Both**

   Explanation:

   - Alnico = Aluminium + Nickel + Cobalt
   - Duralumin = Aluminium + Copper + Magnesium + Manganese
     Both are aluminium-based alloys.

3. **Correct Answer: B) Copper**

   Explanation:
   Copper has very high electrical conductivity and is also ductile, making it ideal for drawing into wires.

4. **Correct Answer: B) Fluorine**

   Explanation:
   Fluorine (in the form of fluoride compounds) helps strengthen tooth enamel and prevent cavities.

5. **Correct Answer: B) Light and corrosion-resistant**

   Explanation:
   Aluminium is lightweight, reducing fuel use, and forms a protective oxide layer that prevents corrosion.

6. **Correct Answer: B) It saves energy and preserves natural resources**

   Explanation:
   Recycling reduces the need for mining and processing ores, which saves energy and natural resources.

7. **Correct Answer: C) Once used, they cannot be easily replaced**

   Explanation:
   Non-renewable resources (like metals and fossil fuels) take millions of years to form, so they must be conserved.

8. **Correct Answer: A) Iron and Carbon**

   Explanation:
   Steel is primarily made by mixing carbon with iron to improve strength and hardness.

9. **Correct Answer: B) Duralumin**

   Explanation:
   Duralumin is strong but very light, making it suitable for aircraft and spacecraft structures.

10. **Correct Answer: A) Iron**

    Explanation:
    Hemoglobin contains iron, which binds oxygen and helps transport it in the blood.

Worksheet-4

1. **Correct Answer: B) Mercury**

   Explanation:

Mercury (Hg) is a very unreactive metal found at the bottom of the reactivity series. Its ore, cinnabar (HgS), can be converted to mercury by simple heating:

1. $2HgS + 3O_2 \rightarrow 2HgO + 2SO_2$
2. $2HgO \rightarrow 2Hg + O_2$

Highly reactive metals like Na, Al, and Ca cannot be obtained by heating alone; they require electrolysis. Iron requires carbon reduction.

2. Correct Answer: B) Electrolysis

Explanation:

- These metals are at the top of the reactivity series. They form very stable compounds, which cannot be reduced by carbon or other reducing agents.
- Electrolysis of their molten salts is required to obtain them.

3. Correct Answer: B) Remove gangue

Explanation:

- Gangue is the unwanted impurity in the ore.
- A flux is added to react chemically with the gangue and form slag, which can be removed easily.
- Acidic gangue (SiO_2) reacts with basic flux (CaO), and basic gangue (FeO) reacts with acidic flux (SiO_2).

4. Correct Answer: B) Minerals

Explanation:

- Minerals are naturally occurring substances in the Earth's crust containing metal compounds.
- If a mineral has a high enough concentration of a metal for economical extraction, it is called an ore.

5. Correct Answer: B) CaO

Explanation:

- Acidic gangue (like SiO_2) is neutralized by a basic flux such as CaO.
- Reaction forms slag ($CaSiO_3$), which separates easily from molten metal.

6. Correct Answer: B) Roasting

Explanation:

- Sulphide ores like PbS are converted to oxides by heating in the presence of oxygen:
  $2PbS + 3O_2 \rightarrow 2PbO + 2SO_2$
- This prepares the ore for reduction to the metal.

7. Correct Answer: B) Copper sulphide, iron, and sulphur

Explanation:

- Copper obtained after reduction of copper pyrites ($CuFeS_2$) is called blister copper.
- It contains residual copper sulphide, iron, and sulphur.
- This requires further refining to obtain pure copper.

8. Correct Answer: B) Combined in one large chamber

Explanation:

- In a blast furnace, both the ore and fuel are placed together in a single large chamber.
- The hearth holds the molten metal, while the fire box produces heat.

9. Correct Answer: B) It covers the metal with another protective metal

Explanation:

- Electroplating deposits a thin layer of less reactive or unreactive metal (like Zn, Sn, or Ni) on the surface of the main metal.
- This prevents corrosion by acting as a protective barrier.

10. Correct Answer: B) A more reactive metal corrodes instead of the main metal

Explanation:

- A sacrificial anode of a more reactive metal (like Mg or Zn) is attached to the main metal.

- The reactive metal corrodes preferentially, protecting the main metal from corrosion.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: B) In disinfecting water**

   Explanation: Chlorine kills harmful microorganisms, so it is widely used to purify drinking water and swimming pools.

2. **Correct Answer: B) It uses less energy than extracting from ore**

   Explanation: Recycling aluminium saves up to ~95% of the energy compared to extracting it from bauxite.

3. **Correct Answer: B) Silver**

   Explanation: Among all metals, silver has the highest electrical conductivity.

4. **Correct Answer: A) Metals**

   Explanation: Metals generally have strong metallic bonds, giving them high melting points compared to non-metals.

5. **Correct Answer: A) Cu, Zn**

   Explanation: Brass is made of copper and zinc.

6. **Correct Answer: D) Copper**

   Explanation: Copper is a less reactive metal and does not react with cold or hot water.

7. **Correct Answer: C) Hydrogen**

   Explanation: Reactive metals form metal hydroxide/oxide and hydrogen gas.

8. **Correct Answer: B) Oxygen**

   Explanation: White phosphorus reacts with oxygen in air and can catch fire, so it is stored under water.

9. **Correct Answer: C) Iron**

   Explanation: Noble metals (Au, Ag, Pt) are least reactive; iron is reactive and rusts.

10. **Correct Answer: A) Mercury**

    Explanation: Mercury is the only metal liquid at room temperature.

11. **Correct Answer: D) Zn (Zinc)**

    Explanation: Zinc coating prevents rusting by acting as a protective and sacrificial layer.

12. **Correct Answer: D) Both A and B**

    Explanation: Brass and bronze are commonly used for making coins.

13. **Correct Answer: D) Potassium**

    Explanation: Potassium reacts violently with air and water, forming a basic solution (KOH) that turns red litmus blue.

14. **Correct Answer: C) Concentrated HCl : Concentrated HNO_3 = 3 : 1**

    Explanation: This mixture can dissolve noble metals like gold and platinum.

15. **Correct Answer: C) Sulphur**

    Explanation: Sulphur forms cross-links between rubber chains, making it stronger and elastic.

16. **Correct Answer: B) Gallium**

    Explanation: Gallium melts at about 29.8°C, close to room temperature.

17. **Correct Answer: C) Semi conductivity**

    Explanation: Silicon is a semiconductor, used in electronic devices.

18. **Correct Answer: C) HNO_3**

    Explanation: Nitric acid is a strong oxidising agent and usually does not liberate hydrogen gas.

19. **Correct Answer: B) Au**

    Explanation: Gold does not react with HCl or HNO_3 alone but dissolves in aqua-regia.

20. **Correct Answer: D) Fe**

    Explanation: Iron does not react with cold or hot water, only with steam.

21. **Correct Answer: C) (B) & (D) → Gold and Silver**

    Explanation: Noble metals like Au and Ag occur in free (native) state.

22. **Correct Answer: D) Hg**

Explanation: Mercury remains liquid at room temperature.

23. Correct Answer: B) Ag

Explanation: Silver is least reactive, so Mg, Zn, and Cu can all displace it from its salt solution.

24. Correct Answer: B) Na (Sodium)

Explanation: Sodium is a soft alkali metal stored in kerosene.

25. Correct Answer: D) Steel

Explanation: Steel = Iron + Carbon (non-metal).

26. Correct Answer: D) Zinc amalgam

Explanation: An amalgam is an alloy of mercury with another metal.

27. Correct Answer: C) Electrolysis of molten compounds

Explanation: Highly reactive metals (Na, K, Ca, Al) cannot be reduced by carbon.

28. Correct Answer: B) Carbon monoxide

Explanation: In the blast furnace, CO reduces iron oxide to iron.
$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

29. Correct Answer: B) Silver > Gold > Copper > Lead

Explanation:

- Sonority is the property of a metal to produce a clear ringing sound when struck.
- Metals that are closely packed, strong, and dense are more sonorous.

30. Correct Answer: B) To reduce zinc oxide

Explanation:

- Zinc is extracted from zinc oxide (ZnO) obtained after roasting zinc sulphide (ZnS).
- The reaction is:
  $ZnO + C \rightarrow Zn\ (vapour) + CO$
- Carbon acts as a reducing agent, removing oxygen from zinc oxide and producing zinc vapour.
- Zinc vapour is then condensed to obtain solid zinc.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

31. Correct Answers: A, B, D

Explanation:
Metals are malleable (can be beaten into sheets), ductile (drawn into wires), and lustrous (shiny). They are good conductors of heat, so option C is incorrect.

32. Correct Answers: A, B, D

Explanation:
Non-metals can be solids (sulphur), liquids (bromine), or gases (oxygen). They generally have low melting and boiling points and are poor conductors of electricity. They are not ductile or malleable, so C is incorrect.

33. Correct Answers: A, B, D

Explanation:
Metals lose electrons to form positive ions (cations), so they act as reducing agents. Many metals corrode when exposed to air and moisture (like rusting of iron). They do not gain electrons, so C is incorrect.

34. Correct Answers: A, C

Explanation:
Duralumin and other aluminium alloys are lightweight, strong, and corrosion-resistant, making them suitable for aircraft. Bronze and brass are heavier.

35. Correct Answers: A, B, C

Explanation:

- Iron → Needed for hemoglobin
- Copper → Helps in enzyme function and iron metabolism

- Zinc → Important for immunity and growth
  Lead is toxic, not essential.

ASSERTION AND REASON TYPE

36. Correct Answer: A)

Explanation:

- Assertion: True — Copper is widely used in wiring.
- Reason: True — Copper is a very good conductor of electricity and is also corrosion-resistant, which makes it suitable for long-term use in wires.
- The reason correctly explains the assertion.

37. Correct Answer: A)

Explanation:

- Assertion: True — Most metals react with acids to form salt and hydrogen gas.
- Reason: True — Metals lose electrons (are oxidized) and form cations, while hydrogen ions from the acid gain electrons and are released as hydrogen gas.
- The reason explains the assertion.

38. Correct Answer: A)

Explanation:

- Assertion: True — Non-metals generally do not conduct electricity (except graphite).
- Reason: True — Their electrons are tightly held and they lack free/delocalized electrons needed for conduction.
- The reason correctly explains the assertion.

39. Correct Answer: A)

Explanation:

- Assertion: True — Steel is generally stronger and harder than pure iron.
- Reason: True — Steel is an alloy of iron with carbon. The presence of carbon improves strength and hardness.
- The reason explains why steel is stronger.

40. Correct Answer: A)

Explanation:

- Assertion: True — Most metals can be beaten into sheets (malleable) and drawn into wires (ductile).
- Reason: True — Delocalized electrons allow layers of metal atoms to slide over each other without breaking bonds.
- The reason correctly explains the assertion.

MATRIX MATCHING TYPE

41.

1. B. Electrical wiringCopper is an excellent conductor of electricity and is highly ductile (can be drawn into wires).

2. A. Aircraft/SpacecraftAluminium is chosen for aerospace because it is low density (lightweight) yet strong when alloyed.

3. C. Musical instrumentsBrass (an alloy of copper and zinc) has acoustic properties and corrosion resistance ideal for horns and trumpets.

4. D. Bridges and machinesSteel is an alloy of iron and carbon known for its high tensile strength and durability in large structures.

42.

1. B. Added to toothpasteFluoride ions help strengthen tooth enamel and prevent cavities.

2. A. Disinfects waterChlorine is a powerful oxidizing agent used to kill bacteria and pathogens in drinking water and pools.

3. C. Rubber vulcanizationSulfur is essential for vulcanization, a process that makes rubber harder and more heat-resistant.

4. D. Fertilizers (superphosphate)Phosphorus is a primary macronutrient for plants, crucial for energy transfer and root development.

COMPREHENSION TYPE

43. Correct Answer: B) Ductility

Explanation: Ductility is the physical property of a material that allows it to be stretched or pulled into a thin wire without breaking. Gold and copper are among the most ductile metals.

44. Correct Answer: B) Hydrogen

Explanation: When a metal reacts with a dilute acid (like Hydrochloric acid), it typically undergoes a displacement reaction. The metal replaces the hydrogen in the acid, releasing Hydrogen gas (H_2) and forming a metal salt.

45. Correct Answer: A) Copper and Zinc

Explanation: Brass is a substitutional alloy made by combining Copper (approx. 66%) and Zinc (approx. 34%). It is highly valued for its gold-like appearance and its resistance to corrosion.

46. Correct Answer: B) Fluorine

Explanation: Fluorine (specifically in the form of fluoride) is added to toothpaste and drinking water because it reacts with the enamel on your teeth. It makes the enamel more resistant to acids produced by bacteria, effectively preventing cavities.

47. Correct Answer: B) Brittle and poor conductors

Explanation: Unlike metals, solid non-metals are brittle, meaning they break or shatter into a powder when struck with a hammer. They are also poor conductors of heat and electricity because they do not have "free" electrons to carry a charge.

48. Correct Answer: B) Fertilizers

Explanation: Phosphorus is one of the three main nutrients (NPK: Nitrogen, Phosphorus, Potassium) required for plant growth. It is primarily used to produce phosphate fertilizers, which help in root development and energy transfer within the plant.

INTEGER TYPE

49. Copper has an atomic number of 29.

Explanation: The atomic number represents the number of protons in the nucleus of an atom. On the periodic table, Copper (Cu) is located in Group 11 and Period 4 with an atomic number of 29.

50. Percentage of iron in steel

Solution: 98%

Explanation: An alloy is a mixture where the total percentage of all components must equal 100%. If steel consists only of iron and carbon for the sake of this calculation:

100% (Total) - 2% (Carbon) = 98% Iron

51. Percentage of zinc in brass

Solution: 30%

Explanation: Just like the steel example, brass is a binary alloy (made of two parts). If the sample is defined by its copper and zinc content:

100% (Total) - 70% (Copper) = 30% Zinc

52. Oxygen constitutes about 46.6% of the Earth's crust by weight.

Explanation: While oxygen is a gas in our atmosphere, in the Earth's crust it is the most abundant element because it is chemically bonded with other elements to form minerals (like silicates and oxides).

Previous year questions

1. Correct Answer: C. III only

Explanation:Reacts with hot waterIron (Fe) does not react with liquid water (hot

or cold). It only reacts with steam.Incorrect

2. Correct Answer: A) I and III only

Explanation:

A displacement reaction occurs only if a more reactive metal is added to a solution of a less reactive metal. The reactivity order here is: Mg > Al > Zn > Fe > Cu.

3. Correct Answer: B) Carbon, magnesium and gold only

Explanation:

Carbon Marked as ductile . This is incorrect; carbon (a non-metal) is brittle and cannot be drawn into wires.

MagnesiumMarked as not malleable and not ductile . This is incorrect; magnesium is a metal and possesses both properties.

GoldMarked as not ductile and not hard . This is incorrect; gold is highly ductile. While pure gold is soft, it is generally categorized as a "hard" solid compared to non-metals.

4. Correct Answer: D) Cu and Ag

Explanation:

If a reaction occurs in all three beakers, the metal strip must be more reactive than the metal in the solution (X or Y).

- Set-up I: Iron (Fe) must be more reactive than X. ($Fe > X$)
- Set-up II: Zinc (Zn) must be more reactive than X. ($Zn > X$)
- Set-up III: Metal X must be more reactive than Y. ($X > Y$)

5. Correct Answer: B) Na, P, Fe

Explanation:

- (i) X is stored under kerosene: Sodium (Na) and Potassium (K) are stored in kerosene because they react vigorously with both air and moisture.
- (ii) Y catches fire on exposure to air and is stored in water: This is a classic property of Phosphorus (P), specifically white phosphorus.
- (iii) Z reacts with steam but not cold/hot water: As explained in Question 1, Iron (Fe) or Zinc (Zn) fits this description.

6. Correct Answer: C) III only

Explanation:

- Beaker I & II: Ca and Mg react with water to release Hydrogen gas (H_2), which burns with a "pop" sound.
- Beaker III: Copper (Cu) does not react with dilute HCl.
- Beaker IV: Fe displaces Zn from $ZnSO_4$ only if it is more reactive; however, Zn is more reactive than Fe, so no reaction occurs.
- Beaker V: Al is more reactive than Zn and will displace it. Since both solutions are colorless, no color change occurs.
- Statement III is the only correct one: Reaction occurs in Beaker II ($Mg + H_2O$), but the solution remains colorless.

7. Correct Answer: A) Colour change will be observed in beakers I, III and IV only

Explanation:

- I: Zinc (Zn) is more reactive than Copper (Cu). The blue $CuSO_4$ solution will fade. (Reaction).
- II: Copper (Cu) is less reactive than Zinc (Zn). (No Reaction).
- III: Zinc (Zn) is more reactive than Iron (Fe). The green $FeSO_4$ solution will fade. (Reaction).
- IV: Iron (Fe) is more reactive than Copper (Cu). The blue $CuSO_4$ will turn light green as $FeSO_4$ forms. (Reaction).

8. Correct Answer:d

i).Copper vessel exposed to moist air

When copper is exposed to moist air (containing water, oxygen, and carbon

dioxide), it undergoes corrosion to form a green coating. This coating is a mixture of copper hydroxide and copper carbonate, which is basic in nature.

- (i) = Green
- Characteristic: Basic substances turn red litmus blue.

ii).Aluminium foil in sodium hydroxide

Metals like aluminium react with bases like sodium hydroxide to produce Hydrogen gas.

- Characteristic: Hydrogen is a colorless, odorless gas that burns with a pop sound.
- (ii) = Pop sound

iii). Rusting of iron

Rusting produces iron oxide, which is a metallic oxide. Most metallic oxides are basic.

- Characteristic: Being basic, it turns red litmus solution blue.
- (iii) = Red litmus solution blue

iv). Burning of sulphur powder

Sulphur is a non-metal. When it burns in air, it forms Sulphur dioxide (SO_2), which is a colourless gas with a suffocating smell.

- (iv) = Colourless, Suffocating
- Characteristic: Non-metallic oxides are acidic, so it turns blue litmus red.

9. Correct Answer: C

The Reaction with Copper Sulphate ($CuSO_4$)

The problem states that when Metal X is placed in copper sulphate, copper falls to the bottom. This means a displacement reaction occurred:

$X + CuSO_4 \longrightarrow XSO_4 + Cu$

- Conclusion: Metal X is more reactive than Copper (Cu).

2. The Reaction with Aluminium Sulphate ($Al_2(SO_4)_3$)

The problem states that when Metal X is placed in aluminium sulphate, no change is observed.

- Conclusion: Metal X is less reactive than Aluminium (Al**)**.

10. Correct Answer: d

V is more reactive than W but less reactive than U:

- From Beaker IV, we know V < U (V is less reactive than U).
- If we combine the logical chain of reactions: if V > X (Beaker III) and $X > W$ (Beaker II), then V must be more reactive than W.

4. Solutions

Solutions

Worksheet – 1

1. **Correct Answer: C**

   A homogeneous mixture of two or more components is called a solution because its composition is uniform throughout.

2. **Correct Answer: A**

   A solution consisting of two components is called a binary solution ("bi" means two).

3. **Correct Answer: A**

   The substance present in relatively lesser quantity in a solution is called the solute; it gets dissolved in the solvent.

4. **Correct Answer: C**

   A solution composed of three components is called a ternary solution ("ter" means three).

5. **Correct Answer: B**

   A true solution has only one phase because it is completely homogeneous throughout.

6. **Correct Answer: B**

   If solute particle size is less than 1 nm, it forms a true solution.

7. **Correct Answer: B**

   A solution of water and sugar is called syrup.

8. **Correct Answer: C**

   In seltzer (carbonated water), the dissolved solute is carbon dioxide (CO_2).

9. **Correct Answer: A**

   A solution of water and salt is called brine.

10. **Correct Answer: C**

    A solution that cannot dissolve more solute at a given temperature is a saturated solution.

Worksheet – 2

1. **Correct Answer: C**

   Solubility is the maximum amount of solute that can be dissolved in a given amount of solvent at a specific temperature.

2. **Correct Answer: B**

   Heating increases the kinetic energy of water molecules, which helps sugar particles dissolve faster in hot tea than in iced tea.

3. **Correct Answer: C**

   The solubility of a gas in a liquid increases when the pressure of the gas above the liquid increases (Henry's law).

4. **Correct Answer: C**

   The SI unit of density is kilogram per cubic meter (kg/m^3).

5. **Correct Answer: C**

   Relative density is a ratio of two densities, so the units cancel out, making it unitless.

6. **Correct Answer: B**

   Mass is measured using a digital weighing balance.

7. **Correct Answer: C**

   For a colourless liquid, the reading should be taken at the bottom of the meniscus.

8. **Correct Answer: B**

   When a substance is heated, it expands, increasing its volume and thus decreasing its density.

9. **Correct Answer: C**

   Gas particles are far apart, so applying pressure compresses them, increasing density.

10. **Correct Answer: B**

    Water has its maximum density at 4°C.

Competitive Corner:

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: C**

   Dissolving glucose in water is a physical change because no new substance is formed.

2. **Correct Answer: C**

   Ethanol in water is a liquid in liquid type solution.

3. **Correct Answer: C**

   Salt dissolves completely in water, forming a homogeneous solution.

4. **Correct Answer: C**

   If the quantity of solute is more, the solution is called a concentrated solution.

5. **Correct Answer: B**

   An unsaturated solution can dissolve more solute at a given temperature.

6. **Correct Answer: A**

   A supersaturated solution contains more solute than required for saturation at a given temperature.

7. **Correct Answer: D**

   Cough syrup is a suspension because solid particles are dispersed in liquid and may settle on standing.

8. **Correct Answer: A**

   The scattering of light by colloidal particles is called the Tyndall effect.

9. **Correct Answer: B**

   If solute particle size is between 1 nm and 100 nm, the solution is a colloid.

10. **Correct Answer: C**

    Salt and water form a true solution, which does not show the Tyndall effect.

11. **Correct Answer: C**

    If the solute particle size is greater than 100 nm, it is called a suspension.

ONE OR MORE THAN ONE CORRECT TYPE:

12. **Correct Answer: B, D**

    Solutions are not pure substances; they are homogeneous mixtures. Preparation of a solution is a physical change, not a chemical change.

13. **Correct Answer: A, C, D**

    Natural water contains dissolved salts and minerals. Dissolved minerals give natural water its taste. Distilled water tastes flat because it lacks dissolved minerals.

14. **Correct Answer: A, B, C**

    Suspensions are heterogeneous mixtures. Their particles are large enough to be seen with the naked eye. Their particles scatter light (show Tyndall effect).

15. **Correct Answer: A, B, C**

    kg/m^3 (SI unit), g/cm^3 and g/mL are the units of the density.

ASSERTION AND REASON TYPE:

16. **Correct Answer: A**

    The solubility of a gas in a liquid is directly proportional to the pressure of the gas above it.

17. **Correct Answer: C**

    Colloidal particles do not settle down, but their size is between 1 nm and 100 nm, not smaller than 1 nm.

18. **Correct Answer: C**

    Drinking only distilled water for a long time is not healthy because it lacks essential minerals. Distilled water does not contain important minerals; it is free from dissolved salts and minerals.

19. **Correct Answer: B**

    Heating decreases density because volume increases, but measuring volume with a measuring cylinder does not explain this decrease.

20. Correct Answer: C

Ice floats because it occupies more volume and is less dense than water, but not all solids are denser than their liquid forms (water is an exception).

MATRIX MATCHING TYPE

21. Correct Answer: a-R, b-S, c-Q, d-P

Binary solution containing 2 components. Ternary solution consists of 3 components while quaternary solution contains 4 components. Dilute solution is a solution that has relatively small amount of solute.

22. Correct Answer: a-Q, b-S, c-P, d-R

Syrup is solution of water and sugar. The solution of water and salt is brine. Selter is carbon dioxide in water. Ammonia water is ammonia in water.

COMPREHENSION TYPE

Passage - I

23. Correct Answer: A

Brine is a solution of common salt (NaCl) in water, so NaCl is the solute.

24. Correct Answer: D

Air is a mixture of gases; nitrogen (solvent) and oxygen (solute) are both gases.

25. Correct Answer: C

Sugar in water forms a true solution.
Milk is a colloid, cheese is not a solution, and sawdust in water forms a suspension.

Passage – II

26. Correct Answer: C

Ice floats on water because it is less dense than water (it occupies more volume for the same mass).

27. Correct Answer: C

Compared to liquid water, ice occupies more volume.

28. Correct Answer: B

Generally, solids are denser than their liquid forms (water is an exception).

INTEGER TYPE

29. Correct Answer: 62.5. Density = Mass ÷ Volume = 500 g ÷ 8 mL = 62.5 g/mL

30. 1700. 1 litre = 1000 mL ⇒ 1.7 × 1000 = 1700 mL

Previous Year Questions

1. Correct Answer: C

Brine and soda water are true solutions, so they do not scatter light and hence do not illuminate the path of light.

Blood and milk are colloids, which scatter light (Tyndall effect), so they illuminate the path of light.

2. Correct Answer: D

When salt is added to water and stirred, it dissolves without significantly increasing the water level. This shows that matter is made up of particles and salt particles occupy the spaces between water particles.

5. Coal and Petroleum

Solutions

Worksheet – 1

1. **Correct Answer: D**
   Petroleum is called black gold because of its dark colour and its high economic value as a major source of fuels and many useful products.
2. **Correct Answer: D**
   Coal is an exhaustible (non-renewable) resource because it is available in limited quantity and takes millions of years to form.
3. **Correct Answer: C**
   Coal is obtained from the earth by digging coal mines, so coal deposits are called mines.
4. **Correct Answer: C**
   On complete combustion, carbon in coal reacts with oxygen to form carbon dioxide (CO_2).
5. **Correct Answer: A**
   Anthracite has the highest carbon content and is the hardest and best quality coal.
6. **Correct Answer: B**
   Exhaustible (non-renewable) resources are available in limited amounts and can be depleted by human use.
7. **Correct Answer: A**
   Sunlight is an inexhaustible (renewable) resource because it is continuously available and does not get used up.
8. **Correct Answer: A**
   Naphthalene is a product separated from coal tar, which is obtained during destructive distillation of coal.
9. **Correct Answer: C**
   When coal is heated in the absence of air, coke is formed. It is black, hard, and porous and contains mostly carbon.
10. **Correct Answer: B**
    Coke is almost pure carbon, whereas coal, coal tar, and coal gas contain many impurities and other substances.

Worksheet – 2

1. **Correct Answer: A**
   Petroleum is formed from the remains of tiny plants and animals living in seas (undersea organisms) that were buried under sediments and converted into petroleum over millions of years.
2. **Correct Answer: A**
   Liquefied form of dead organic matter is known as crude oil.
3. **Correct Answer: B**
   Bitumen is a heavy fraction obtained from petroleum refining and is used for surfacing (metalling) roads.
4. **Correct Answer: C**
   LPG and kerosene are used as cooking fuels, and CNG can also be used as a domestic fuel.
   Petrol is mainly used as a motor fuel, not for cooking.
5. **Correct Answer: C**
   Petroleum refining is the process of separating crude oil into useful fractions (petrol, kerosene, diesel, etc.) by fractional distillation.
6. **Correct Answer: D**
   Crude oil is formed from dead marine organisms buried deep under the earth and converted into petroleum due to high temperature and high pressure over millions of years.
7. **Correct Answer: C**
   The word petroleum comes from Latin: “petra” = rock and “oleum” = oil, so it means rock oil.

8. **Correct Answer: C**
White petrol is used as a solvent in dry cleaning to remove grease and stains from clothes.
9. **Correct Answer: B**
Natural gas is found along with petroleum in oil fields and is often obtained during drilling.
10. **Correct Answer: C**
Aviation turbine fuel (ATF) used in jet aircraft is a refined form of kerosene.

Competitive Corner:

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: D**
Petroleum wax (paraffin wax) is used in making Vaseline, candles, ointments, and polishes.
2. **Correct Answer: C**
Natural gas is used to produce ammonia, which is a raw material for making fertilizers (urea, ammonium salts).
3. **Correct Answer: D**
Paraffin wax is semi-solid at room temperature, whereas petrol and diesel are liquids and coal is solid.
4. **Correct Answer: A**
CNG (Compressed Natural Gas) burns more completely and produces less smoke and harmful gases compared to petrol and diesel.
5. **Correct Answer: C**
Carbonization is the slow process by which dead vegetation get converted into coal under high temperature and pressure over millions of years.
6. **Correct Answer:**
C. Plastic is a man-made (synthetic) material, whereas soil and minerals are obtained directly from nature.
7. **Correct Answer:**
C. Coal and petroleum are formed from the dead remains of plants and animals buried under the earth for millions of years.
8. **Correct Answer:**
B. Coal is formed mainly from dead plants (vegetation) that were buried and converted into coal by heat and pressure.
9. **Correct Answer:**
B. PCRA is an Indian organization that promotes conservation of petroleum products and awareness about fuel saving.
10. **Correct Answer: D**
Earlier, coal gas was used for street lighting in London before the use of electricity became common.
11. **Correct Answer: D**
Coal tar is obtained from coal (during destructive distillation), not from petroleum deposits. LPG, petrol, and kerosene are petroleum products.
12. **Correct Answer: A**
Fossil fuels like coal are obtained from the earth by mining (petroleum and natural gas are obtained by drilling).
13. **Correct Answer: A**
Coal tar is a thick black liquid with an unpleasant smell, whereas coke, coal gas, and petrol do not have such a strong foul odour.
14. **Correct Answer: B**
Burning fossil fuels releases smoke and harmful gases, which cause air pollution and environmental problems.
15. **Correct Answer: A**
The separation of petroleum into different useful fractions is done by refining (fractional distillation) in a refinery.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

16. **Correct Answer: A, B, C**
Natural gas *can* be used directly in homes Natural gas is easily transported through pipelines. Natural gas is used for power generation in thermal power plants.
17. **Correct Answer: A, D**
Methane (main component of natural gas) is used as a fuel in boilers. CNG

means Compressed (confined) Natural Gas.

18. **Correct Answer: A, D**
Petroleum and minerals are exhaustible resources.

19. **Correct Answer: B**
A fossil is the preserved remains or traces of ancient plants or animals found in rocks.

20. **Correct Answer: A, C, D**
Cow dung cakes are commonly used as a fuel in villages. During festivals, oil lamps (using ghee or cooking oil) are used for lighting. Vegetation fuel (firewood, crop waste) is obtained from plants (flora).

ASSERTION AND REASON TYPE:

21. **Correct Answer: A**
Coal and petroleum are formed from the fossilized remains of plants and animals (living organisms) over millions of years, so they are called fossil fuels.

22. **Correct Answer: C**
Coal gas is indeed a mixture of hydrocarbons and carbon monoxide. However, it is used as a fuel and as a source of heat energy, not only for lighting.

23. **Correct Answer: B**
Both statements are true. However, the reason talks about biodegradability and engine compatibility, which does not explain why bio-diesel is produced from biological sources or used as an alternative fuel.

24. **Correct Answer: D**
Petroleum is not a pure substance; it is a mixture of many hydrocarbons. The reason correctly states that petroleum is a mixture of hydrocarbons obtained from deep inside the Earth.

MATRIX MATCHING TYPE

25. **Correct Answer: a→Q, b→P, c→R, d→S**
During the fractional distillation of petroleum, different fractions condense at different temperatures according to their weights. Fuel gas, being the lightest fraction, condenses at temperatures below about 40 °C. Kerosene condenses at a moderate temperature of around 150-250 °C. Diesel oil, which is a heavier fraction, condenses at about 250-350 °C. The heaviest fractions such as lubricating oil and fuel oil condense at very high temperatures, around 350 °C and above.

26. **Correct Answer: a→Q, b→S, c→R, d→P**
Coke is used in the manufacture of steel and for the extraction of metals, as it acts as both a fuel and a reducing agent in blast furnaces. Coal is widely used as a fuel in thermal power plants to produce electricity. Coal tar is a valuable source of many chemicals and is used to make drugs, dyes, naphthalene, and for surfacing roads. Natural gas is a clean fuel commonly used for cooking and heating in homes.

COMPREHENSION TYPE

Passage - I

27. **Correct Answer: A**
Coal on heating without oxygen gives coke, coal tar and coal gas. Since Y is the industrial fuel (coke), X corresponds to coal tar.

28. **Correct Answer:B**
Coal tar is a thick black liquid with an unpleasant smell.

29. **Correct Answer: C**
Coke is used as an industrial fuel and in the manufacture of steel and extraction of metals, so Y is coke.

Passage – II

30. **Correct Answer: D**
Petroleum (crude oil) is a thick, black liquid fossil fuel formed from the buried remains of marine (aquatic) organisms.

31. **Correct Answer: D**
The slow natural process by which dead aquatic organisms get converted into petroleum is called fossilization.

32. **Correct Answer: C**

When petroleum is processed (refined), it produces useful products such as petrol, diesel, kerosene, LPG, etc. Hence product X is petrol.

INTEGER TYPE

33. Correct Answer: 200

Coal tar is a mixture of around 200 substances.

34. Correct Answer: 2

Based on the availability in nature, natural resources are classified into 2 groups. They are exhaustible and inexhaustible natural resources.

35. Correct Answer: 40-120 °C

During fractional distillation of crude oil, petrol (gasoline) has a relatively low boiling point range compared to heavier fractions. Therefore, it condenses and is collected in the upper part of the distillation column at about 40-120 °C.

Previous Year Questions

1. Correct Answer: C

Coke is obtained by carbonisation of coal (heating in absence of air). Coke is produced during destructive distillation of coal. Bitumen is obtained as a residue during petroleum refining. Crude oil is formed from dead organisms buried under the earth over millions of years.

2. Correct Answer: B

W is kerosene. Kerosene is used as aviation fuel. X is coal tar, a black liquid from which naphthalene balls are obtained. Y is CNG, a cleaner fuel used for power generation. Z is bitumen, which is used for surfacing road.

3. Correct Answer: A

The process shown in the given figure is called fractional distillation.
The difference in the (i) boiling point of various components of petroleum is used to separate them. The fraction (ii) diesel will be condensed before fraction (iii) kerosene, as it has (iv) higher boiling point.

4. Correct Answer: A

Petrol condenses near the top of the fractionating column because it has a lower boiling point than diesel. Petrol is more volatile than diesel (i.e., it vaporizes more easily and has a lower boiling point). Statement 2 correctly explains Statement 1, since higher volatility (lower boiling point) makes petrol rise higher and condense near the top.

5. Coal and Petroleum B

Fractions arranged in decreasing order of boiling points:

Heavy oil (ii) > Diesel (iv) > Kerosene (iii) > Petrol (i) > Petroleum gas (v)

6. Coal and Petroleum D

Non-renewable resources cannot be replaced within a reasonable period of time. Coal and natural gas are exhaustible (non-renewable) resources, not inexhaustible. CNG is less polluting than petrol. High temperature and pressure convert buried vegetation into coal over millions of years.

7. Coal and Petroleum A

Producer gas is a mixture of carbon monoxide and hydrogen.

8. Coal and Petroleum D

Bitumen is used for surfacing (metaling) roads. Paraffin wax is used to make candles, ointments, and Vaseline. Petroleum gives petrol, diesel etc., used as motor fuels. Petroleum is called “black gold”, and petrol is a product of it.

9. Coal and Petroleum B

X is coal. Y is coke. Z is coal tar.

6. Combustion and Flame

Solutions

Worksheet – 1

1. **Correct Answer: C**
   Coal reacts slowly with oxygen at ordinary temperature. The heat produced accumulates and may cause the coal to catch fire on its own. This is called spontaneous combustion.
2. **Correct Answer: B**
   White phosphorus has a very low ignition temperature. It catches fire on its own in air at room temperature without external heating, so it is spontaneous combustion.
3. **Correct Answer: A**
   LPG burns quickly when ignited and produces heat and light immediately. This type of combustion is called rapid combustion.
4. **Correct Answer: C**
   Crackers burn suddenly with the production of heat, light, sound, and large amounts of gases. This is called explosive combustion.
5. **Correct Answer: C**
   Coal dust particles burn very rapidly when mixed with air, producing a sudden explosion. Hence, it is explosive combustion.
6. **Correct Answer: B**
   In respiration, food reacts with oxygen to release energy. Since a new substance is formed, it is a chemical process.
7. **Correct Answer: C**
   White phosphorus has a very low ignition temperature and catches fire on its own in air. Others do not burn spontaneously.
8. **Correct Answer: C**
   Asbestos does not burn; it is used as a fire-resistant material. Alcohol, hydrogen, and chaff can burn.
9. **Correct Answer: B**
   Spirit (alcohol) catches fire more easily than kerosene, diesel, or mustard oil because it has a lower ignition temperature.
10. **Correct Answer: C**
    In the Sun, hydrogen produces heat and light by nuclear fusion, not by combustion (chemical reaction with oxygen).

Worksheet – 2

1. **Correct Answer: B**
   White phosphorus is highly reactive and poisonous. Modern safety matches use red phosphorus instead, which is safer.
2. **Correct Answer: D**
   LPG has a calorific value around 40,000–50,000 kJ/kg, which matches the given value better than biogas or methane.
3. **Correct Answer: D**
   Hydrogen has the highest calorific value (about 150,000 kJ/kg), much higher than natural gas, LPG, or coal gas.
4. **Correct Answer: B**
   Extreme heat, lightning, or even a small spark can ignite dry grass and start a forest fire.
5. **Correct Answer: C**
   CO_2 does not conduct electricity and leaves no residue, so it is safe for electrical fires.
6. **Correct Answer: D**
   Incomplete burning of coal produces carbon monoxide, a poisonous gas. It reduces oxygen supply in the blood and can cause death.
7. **Correct Answer: B**
   Coke is almost pure carbon. It burns by glowing (no vapor formation), so it does not produce a flame. Substances that vaporize (like kerosene and LPG) burn with a flame.

8. **Correct Answer: C**

Biogas has a lower calorific value (about 35,000 kJ/kg or less) compared to kerosene, CNG, and LPG. Hence, it produces less heat per kilogram of fuel.

9. **Correct Answer: C**

The outermost zone gets plenty of oxygen, so complete combustion takes place there. Hence, it is the hottest part of the flame

10. **Correct Answer: C**

Like a candle flame, a kerosene lamp flame has three zones

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: C**
Carbon dioxide is the main greenhouse gas responsible for global warming due to excessive burning of fossil fuels.
2. **Correct Answer: B**
Acid rain is mainly caused by sulphur dioxide and nitrogen oxides. Carbon monoxide does not form acids in rainwater.
3. **Correct Answer: C**
Natural gas (CNG) burns more completely and produces fewer harmful gases and pollutants compared to petrol and diesel.
4. **Correct Answer: B**
Fire is a form of combustion, which is a rapid chemical reaction that produces heat, light, and smoke.
5. **Correct Answer: A**
Calorific value = Heat energy produced ÷ Amount of fuel taken
= 33,000 ÷ 6 = 5,500 kJ/kg.
6. **Correct Answer: C**
Charcoal does not vaporize during combustion, so it burns without producing a flame.
7. **Correct Answer: C**
The rubbing surface of a safety matchbox contains powdered glass and a small amount of red phosphorus, which is safer than white phosphorus.
8. **Correct Answer: C**
Oxygen is essential for combustion. When the jar is placed over the candle, the available oxygen decreases. Once it falls below the required level to support burning, the flame goes out.
9. **Correct Answer: A**
Paper, matchstick, and straw burn easily (combustible). Stone and sand do not burn (non-combustible).

ONE OR MORE THAN ONE CORRECT OPTION TYPE:

10. **Correct Answer: B, C**
Carbon dioxide is heavier than air and does not support combustion. That is why it is used to extinguish electrical fires.
11. **Correct Answer: B, C**
Both sodium bicarbonate and potassium bicarbonate are used in dry powder fire extinguishers. On heating, they release carbon dioxide, which cuts off oxygen and extinguishes the fire.
12. **Correct Answer: A, D**
Fire needs oxygen to burn. Throwing sand or covering with a woollen blanket cuts off oxygen and extinguishes the fire. Polyester blanket is unsafe because it can melt and stick to the skin.
13. **Correct Answer: B, D**
Green leaves have high moisture content, which absorbs heat and prevents them from reaching ignition temperature. Also, when piled in a heap, they do not get enough oxygen for proper burning.
14. **Correct Answer: A, C**
Wood and kerosene burn in air, so they are combustible. Iron nail and glass do not burn under normal conditions, so they are non-combustible.
15. **Correct Answer: B, D**

Methane and compressed natural gas (CNG) are gaseous fuels. Petrol is a liquid fuel. Charcoal is a solid fuel.

ASSERTION AND REASON TYPE:

16. Correct Answer: A

Sodium and potassium are highly reactive metals. When exposed to air, they react vigorously with oxygen and moisture and may catch fire (spontaneous combustion). Hence, they are stored in kerosene to prevent contact with air and moisture.

17. Correct Answer: D

The luminous zone is not the hottest part of a candle flame; the outer non-luminous zone is the hottest. The luminous zone (~1200°C) is a zone of incomplete combustion, so the Reason is true but does not support the Assertion.

18. Correct Answer: A

Burning fuels produce carbon dioxide, a greenhouse gas. Increased CO_2 concentration enhances the greenhouse effect, leading to global warming. Hence, the Reason correctly explains the Assertion.

19. Correct Answer: B

Respiration occurs at body temperature (about 37°C) because it is controlled by enzymes.
Respiration is a slow process. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

20. Correct Answer: C

Oxygen supports combustion, so the Assertion is true. A candle cannot continue to burn in the absence of air (oxygen is required), so the Reason is false.

MATRIX MATCHING TYPE:

21. Correct Answer: a-Q, b-S, c-P, d-R.

Wood has a calorific value of 17000–22000 kJ/kg. The calorific value of coal is 25000–33000 kJ/kg. Hydrogen's calorific value is 150000. LPG has a calorific value of 55,000 kJ/kg.

22. Correct Answer: a-Q, b-P, c-S, d-R

Wood is a solid substance used as fuel. Petrol is a liquid fuel used in vehicles. Iron nail does not burn easily, so it is non-combustible. Hydrogen is a fuel that exists in gaseous state.

COMPREHENSION TYPE:

Passage – I

23. Correct Answer: C

Combustion is a chemical reaction in which a substance reacts with oxygen to produce heat and light. Hence, it is a process of oxidation.

24. Correct Answer: B

Combustion releases energy mainly as heat and light (e.g., burning candle or LPG flame).

25. Correct Answer: C

Combustibility is the ability of a substance to burn or undergo combustion.
Passage – II

26. Correct Answer: B

Turning off cooking stoves after use prevents gas leakage and accidental fires.

27. Correct Answer: B

Proper ventilation allows fresh air circulation and prevents the accumulation of harmful gases, reducing fire risk.

28. Correct Answer: C

In case of a fire emergency, contacting the fire station quickly helps control the situation and prevent damage.

INTEGER TYPE

29. Correct Answer: 150000.

Hydrogen has the highest calorific value among common fuels, about 150,000 kJ/kg, meaning it produces a large amount of heat per kilogram.

30. Correct Answer: 600

The temperature in non-combustible zone is around 600 °C.

31. Correct Answer: 2941.

Calorific value = 5000 ÷ 1.7 = 2941.17 kJ/kg ≈ 2941 kJ/kg.

Previous Year Questions:

1. **Correct Answer: B.**
   The middle zone of a flame is the luminous zone where incomplete combustion occurs, but generally no residue is left on an object placed there (soot may deposit only in specific cases). With sufficient oxygen, complete combustion occurs and produces a blue flame. In the dark zone, wax vapours are present but have not yet started burning because they are not heated enough, not just because of lack of oxygen. LPG has a higher calorific value than kerosene.
2. **Correct Answer: B**
   Incomplete combustion occurs in the luminous (yellow) zone, not in the non-luminous zone.
3. **Correct Answer: A**
   Burning of phosphorus in air at room temperature is spontaneous combustion, not rapid combustion. Water cools the combustible material by bringing its temperature below ignition temperature, not above it.
4. **Correct Answer: B**
   The innermost zone contains unburnt vapours; combustion does not occur here. Rusting is a very slow oxidation process. LPG burns quickly producing heat and light. Complete combustion occurs in the outer non-luminous (blue) zone.
5. **Correct Answer: D**
   Petrol, Coal, Camphor burn easily and produce heat and light. Glass, sand, water do not burn.
6. **Correct Answer: B**
   Figure (I) shows the outermost zone of the flame, which is the hottest zone. Figure (II) represents the luminous zone, which contains unburnt carbon particles.
7. **Correct Answer: A**
   Efficiency depends on calorific value (higher value → more heat produced per kg).
   Therefore, the correct order of efficiency is: LPG > CNG > Diesel > Coal.
8. **Correct Answer: C**
   A substance cannot burn below its ignition temperature. A matchstick head contains antimony trisulphide and potassium chlorate, not arsenic trisulphide and potassium chloride. Burning of white phosphorus at room temperature is spontaneous combustion, not rapid combustion. Kerosene has a higher calorific value than coal.
9. **Correct Answer: C**
   Because of the presence of water, the temperature of the paper does not rise to its ignition temperature, so the paper does not burn. Instead, the water absorbs the heat and starts boiling. The empty cup burns because its temperature quickly reaches the ignition temperature.
10. **Correct Answer: B**
    The outermost zone of a flame is the hottest zone because complete combustion occurs there. Goldsmiths use this zone to melt silver and gold as it provides maximum heat.